

Brent Hu

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EDUCATION

Yale University

New Haven, CT

Bachelor of Science in Computer Science | GPA 3.92

Expected 2029

Relevant Coursework: Algorithms, Intermediate ML, Data Structures, Discrete Math, Linear Algebra

Awards: AIME Qualifier, USNCO Finalist, Yale College First-Year Summer Research Fellowship

EXPERIENCE

Qernelzoo and Batista Lab

New Haven, CT

AI Researcher

May 2026 – Present

- Built agent harness and engineered prompts for open source models to autonomously calculate environment-dependent pseudopotentials for novel materials with Hugging Face Transformers.
- Integrated custom tools for fetching crystallography databases and querying technical docs with semantic search for Retrieval-Augmented Generation, **reducing factual hallucinations by 94%**.
- Leveraged NERSC Perlmutter supercomputer for large-scale parallel workloads, using SLURM for job scheduling and CPU/GPU resource allocation.

Yale Computer Society: CourseTable Development Team

New Haven, CT

Full-stack Developer

Sep. 2025 – Present

- Built new data ingestion workflows to scrape course data and analytics from university websites, parse and normalize heterogeneous formats, and load structured data into PostgreSQL.
- Parallelized asynchronous HTTP requests in scraping pipelines, **reducing runtime by 80%**.
- Improved CourseTable UI by identifying pain points through student feedback and implementing targeted quality-of-life features for 6,000+ users.

Yale Undergraduate Aerospace Association: Jet Team

New Haven, CT

Controls Software Developer

Sep. 2025 – Present

- Wrote control software for an RC plane, integrating real-time telemetry streams with pilot input.
- Modified open-source ECU firmware to support custom TurboJet engine hardware.

University of Puget Sound

Tacoma, WA

AI Research Assistant

May 2024 – Sep. 2024

- Developed physics-informed neural networks for heat transfer simulation using TensorFlow.
- Optimized custom loss functions and hyperparameters, **dropping convergence time by 56%**.
- Engineered preprocessing pipelines to filter noisy measurements and normalize input data, further improving stability and convergence during neural network training.

PROJECTS

Automated Daily Stock Screener | *Python, Pandas, TradeStation API*

- Integrated TradeStation REST API to automate market-data retrieval across thousands of equities, handling authentication, JSON parsing, and rate limits.
- Implemented configurable backtesting engine to process 10+ years of historical market data, enabling automated strategy evaluation and parameter optimization.

whitepass-patrol-dev.web.app | *React, Material UI, Firebase*

- Deployed scheduling platform supporting 200+ users for White Pass Ski Patrol.
- Wrote event-driven Cloud Functions to automatically maintain denormalized aggregate data, reducing Firestore read operations for frequently queried statistics and associated database costs.
- Gathered user feedback through interviews and usability testing to deliver product improvements.

github.com/brtboi/tractor | *React, SCSS, Express.js, Socket.io*

- Built full-stack implementation of the Tractor card game for low-latency multiplayer gameplay.
- Designed real-time client-server architecture using WebSockets to synchronize game state and player actions, enabling consistent multiplayer interactions across multiple connected clients.

SKILLS

Languages: Python, TypeScript, HTML/CSS, SQL, C, R, Java, Bash, Racket

Frameworks: React, Express.js, Node.js, TensorFlow, Keras, PyTorch, Hugging Face Transformers

Developer Tools: Git, Docker, Linux, Firebase, SLURM, R Studio